

Agent Behavior Analysis

Consolidated Report

Slider Validation | P&L Correlations | Strategy Alignment

Data: January 8-11, 2026

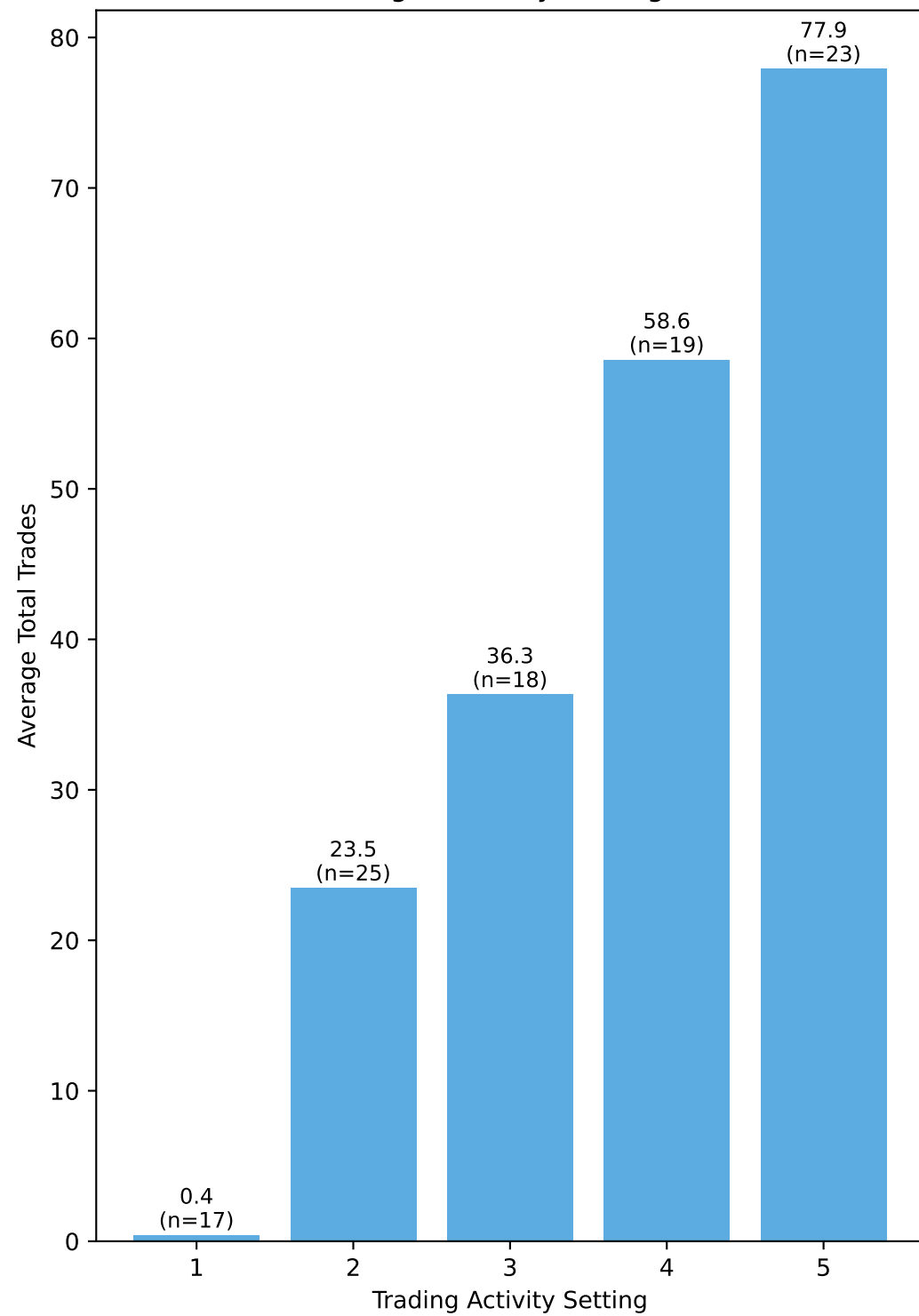
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Key Findings:

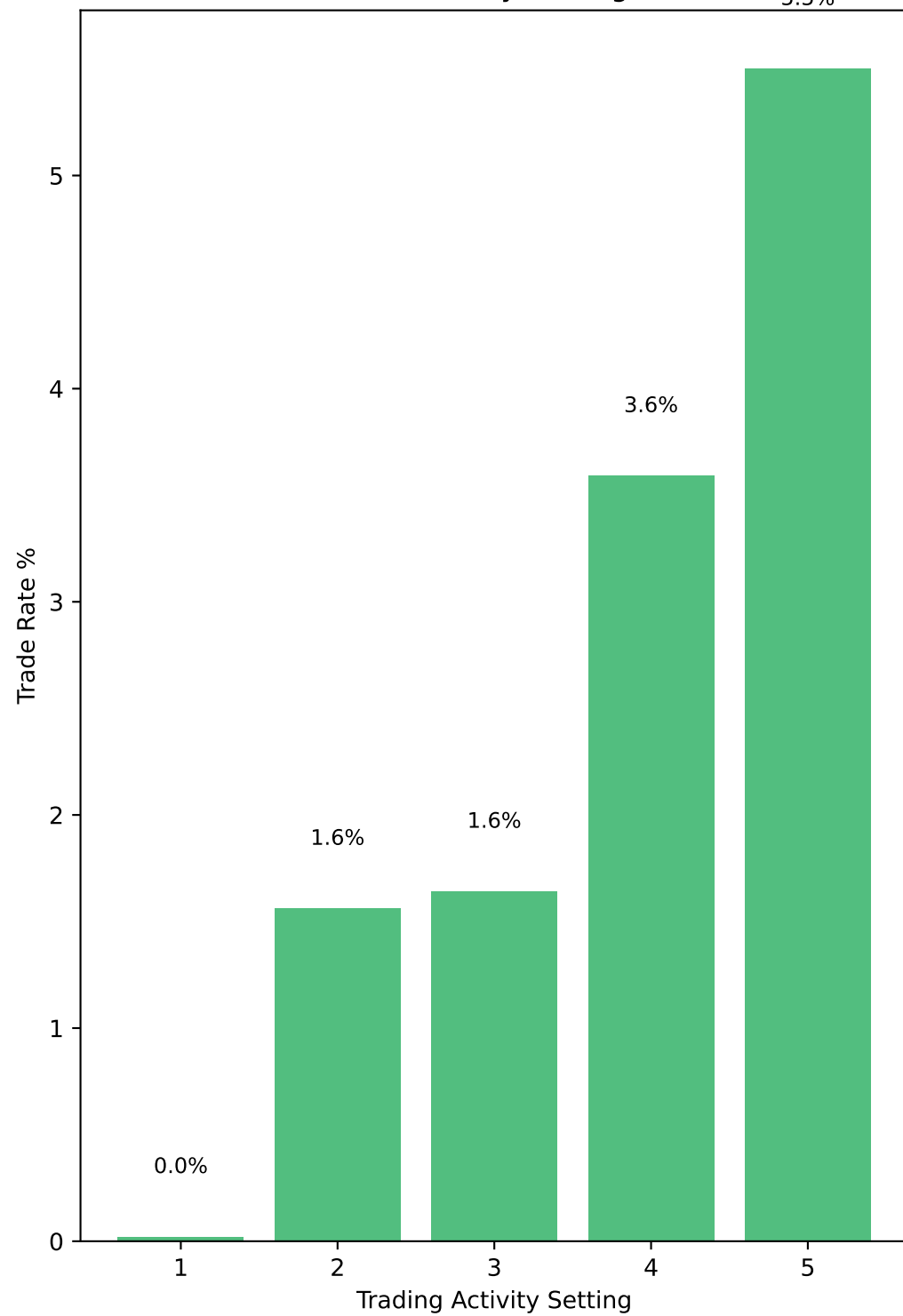
- 72% of trades align with strategies
- Risk preference strongly predicts P&L ($r=-0.51$)
- 93% of trades concentrated in 5 tokens (herding)
- Past behavior predicts future trades (ML: 70% accuracy)
- 73% of agents lost ETH over time (ETH→Token conversion)

Trading Activity Slider Validation (Agents with >1 ETH, n=102)

Avg Trades by Setting

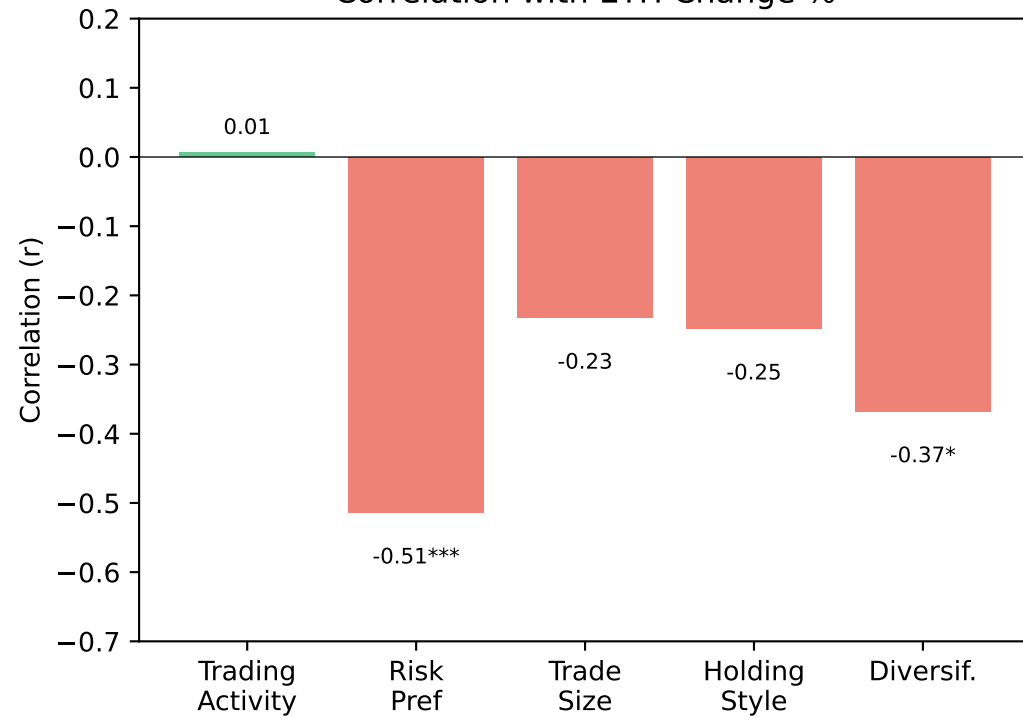


Trade % by Setting



Correlation Analysis: Slider Settings vs Outcomes

Correlation with ETH Change %



Correlation with Total Trades



Correlation with Unique Tokens



SIGNIFICANCE:

*** $p < 0.01$

** $p < 0.05$

* $p < 0.10$

KEY FINDING:

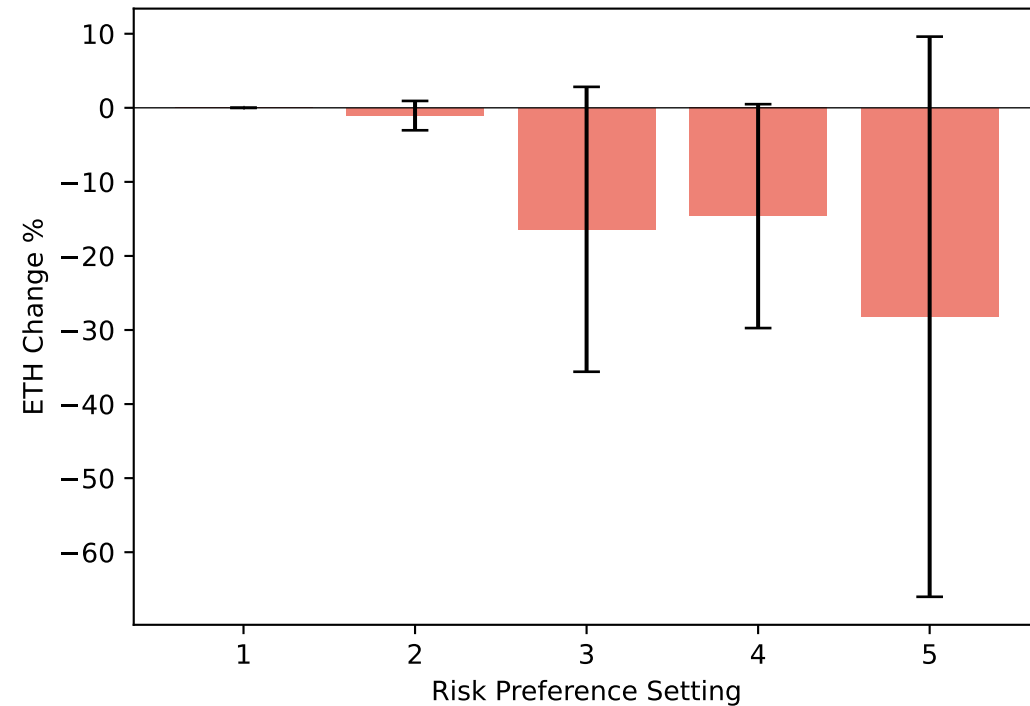
Risk Preference

$r = -0.51$ ($p = 0.006$)***

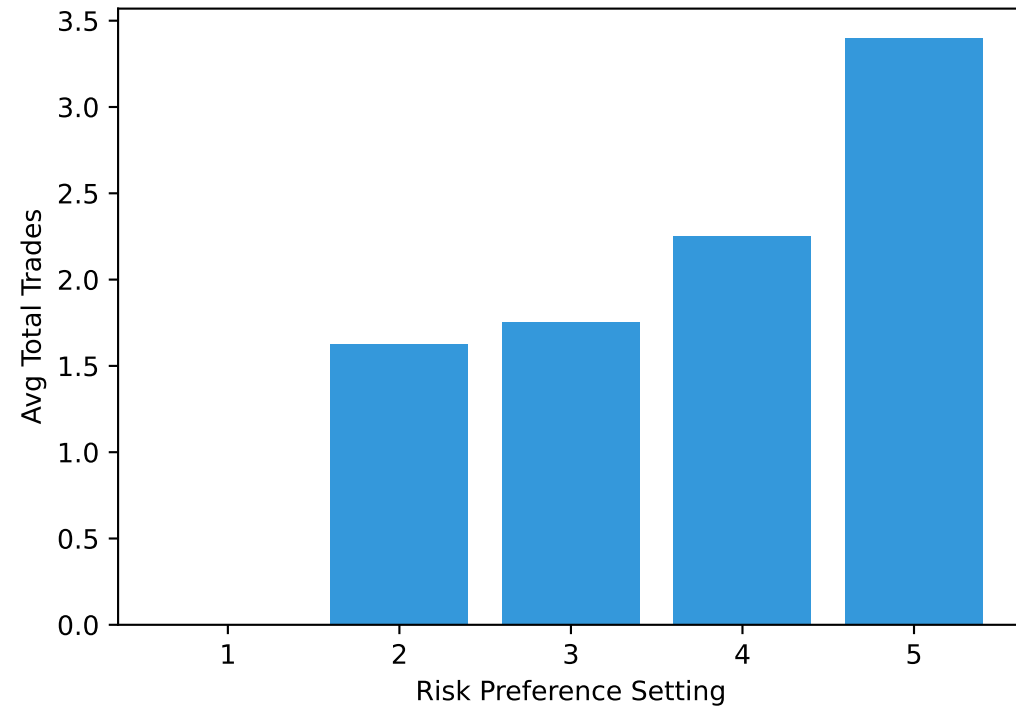
Higher risk = Worse P&L

Risk Preference: The Key P&L Driver

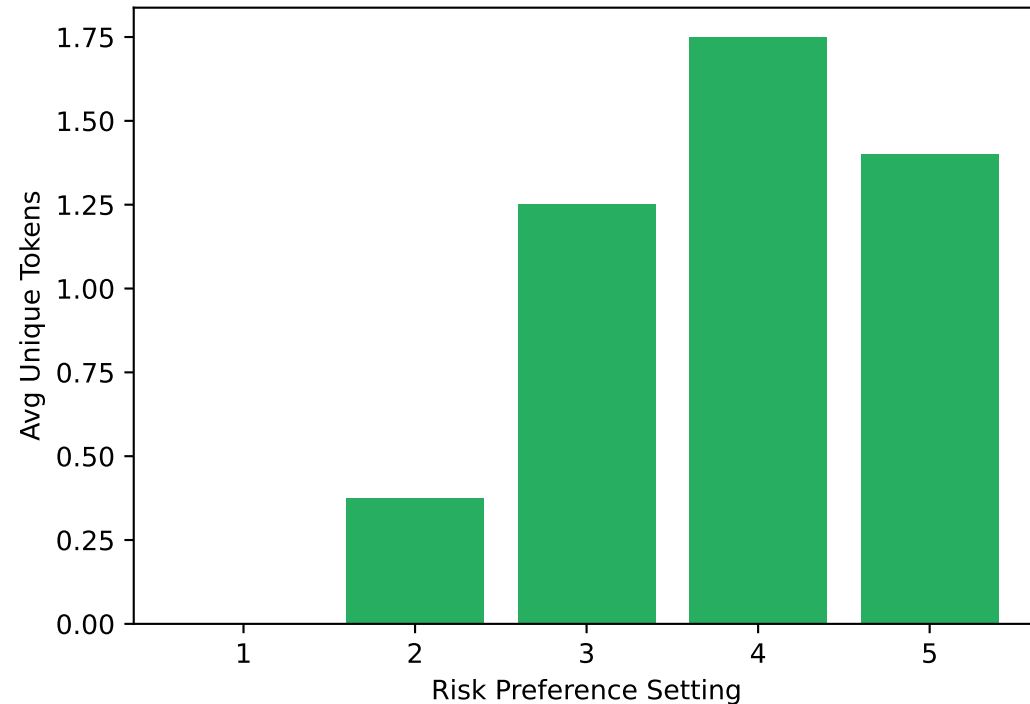
Average P&L by Risk Preference



Trading Activity by Risk Preference



Token Diversity by Risk Preference



RISK PREFERENCE IMPACT:

Setting 1 (Low Risk):

0% avg P&L, 0 trades

Don't trade = Don't lose

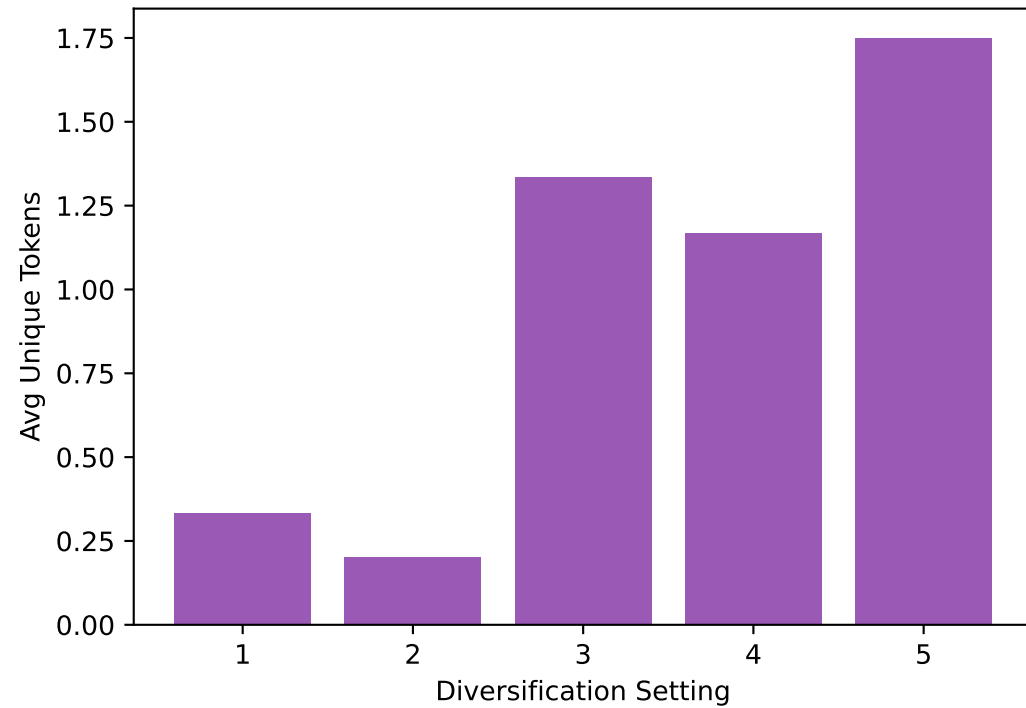
Setting 5 (High Risk):

-28% avg P&L, 3.4 trades

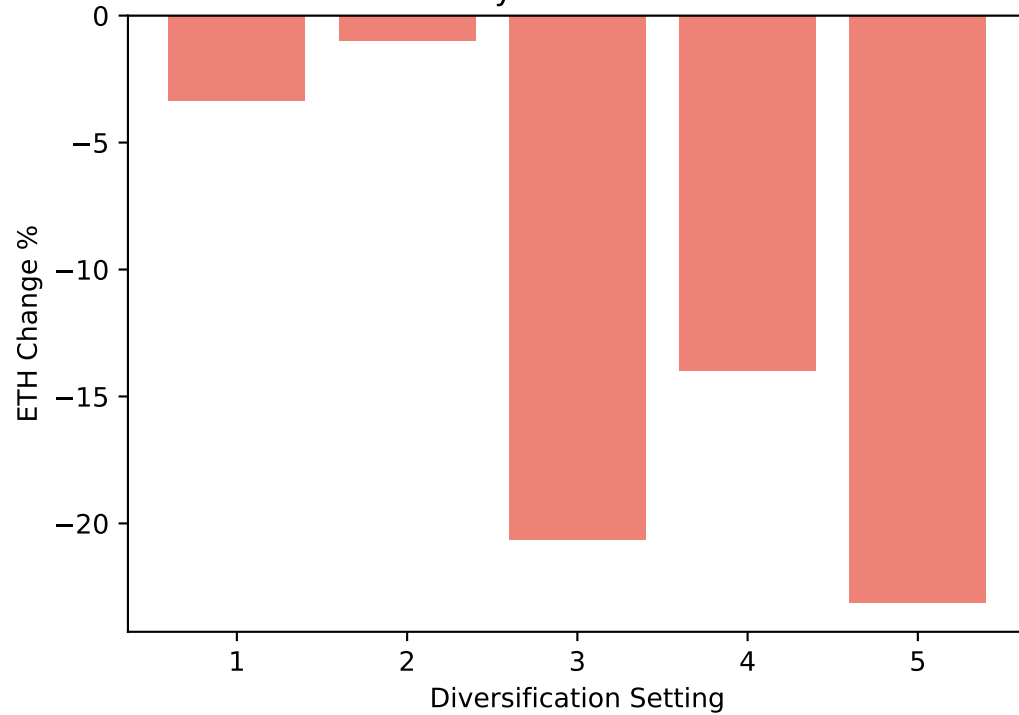
High activity = High losses

Diversification Setting Validation

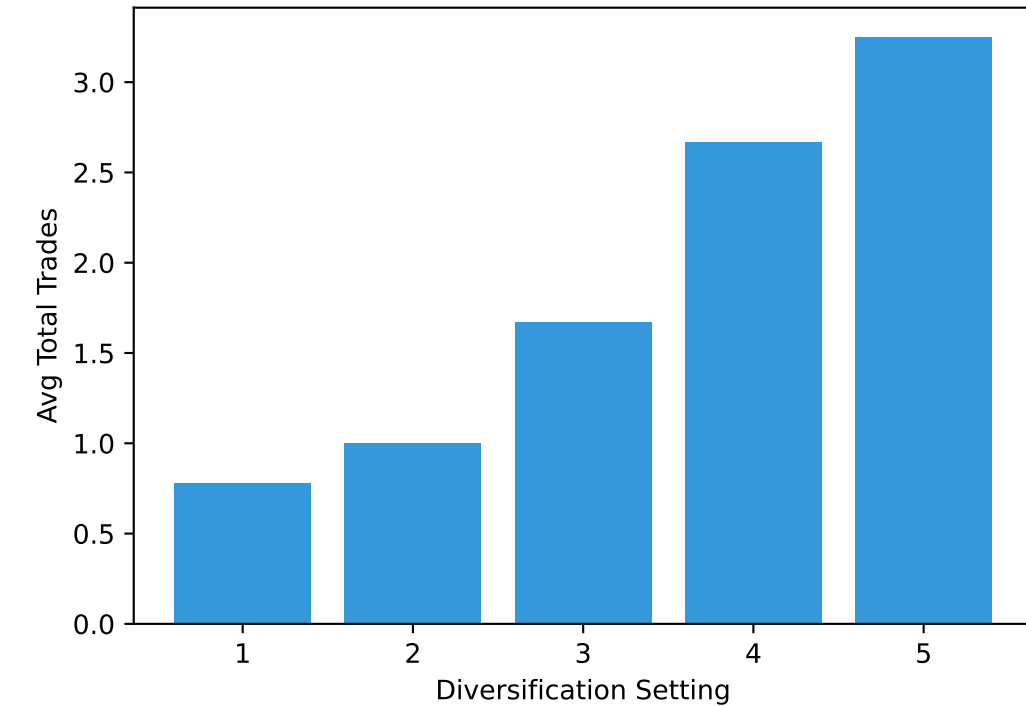
Actual Token Diversity by Setting



P&L by Diversification



Trading Activity by Diversification



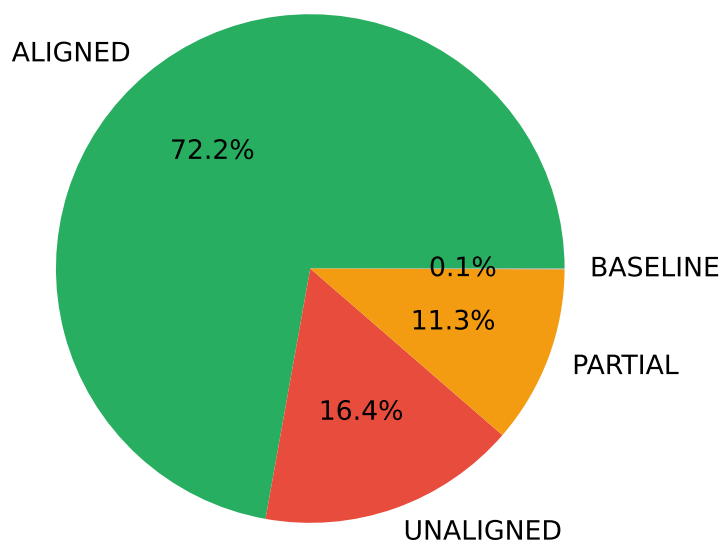
VALIDATION:
Higher setting = More tokens

Setting 1: 0.33 tokens
Setting 5: 1.75 tokens

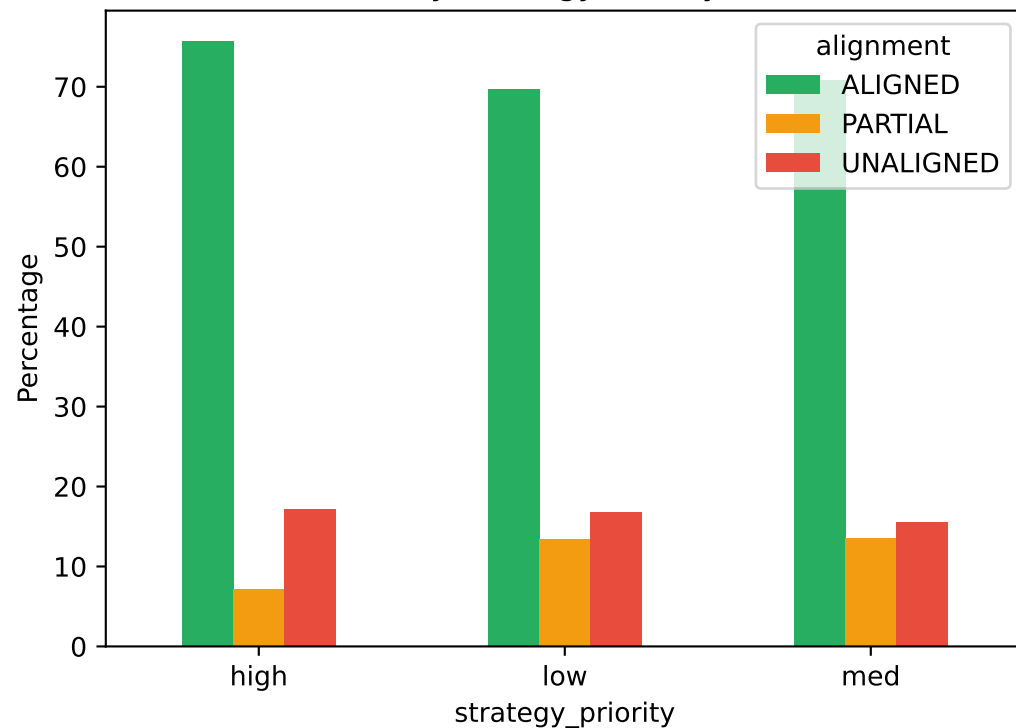
Correlation: $r=+0.47^{**}$

Strategy Alignment Analysis (72% Overall)

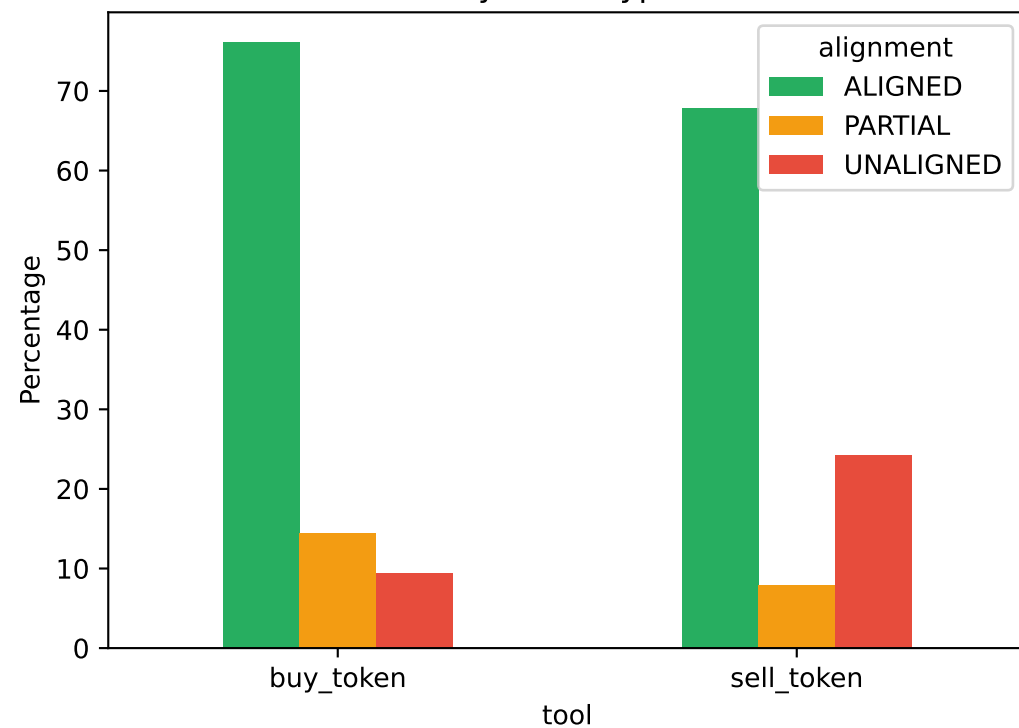
Overall Alignment



By Strategy Priority



By Trade Type



Total trades: 5,903
Unique agents: 306
Unique strategies: 150

Which Strategy Types Get Followed?

ALIGNMENT RATE BY STRATEGY CATEGORY

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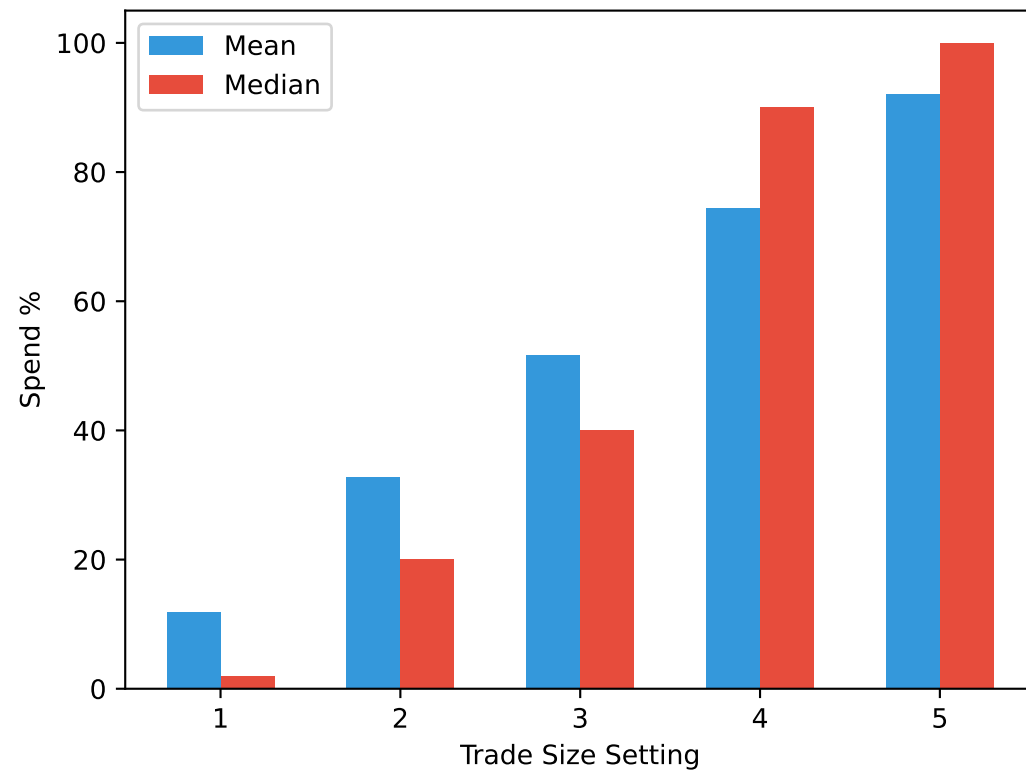
Category	Alignment	Description
MOMENTUM	85.5%	"buy strength after breakout..."
VOLUME_BASED	78.9%	"only trade with rising volume..."
HOLD	78.7%	Strategies that emphasize holding
MULTI_TOKEN	78.3%	Mention 3+ specific tokens
DIP_BUY	76.9%	"buy dips", "mean reversion"
HAS_PRICE_TARGET	75.7%	Include 2x, 3x, etc. targets
DIVERSIFY	68.0%	"diversify equally..."
RESTRICTIVE	66.7%	Use "only" or "ignore"
SINGLE_TOKEN	57.2%	Focus on one token only
LEVERAGE	47.0%	Mention leverage (can't execute!)
ALL_IN	25.5%	"Go all-in" strategies

KEY INSIGHTS:

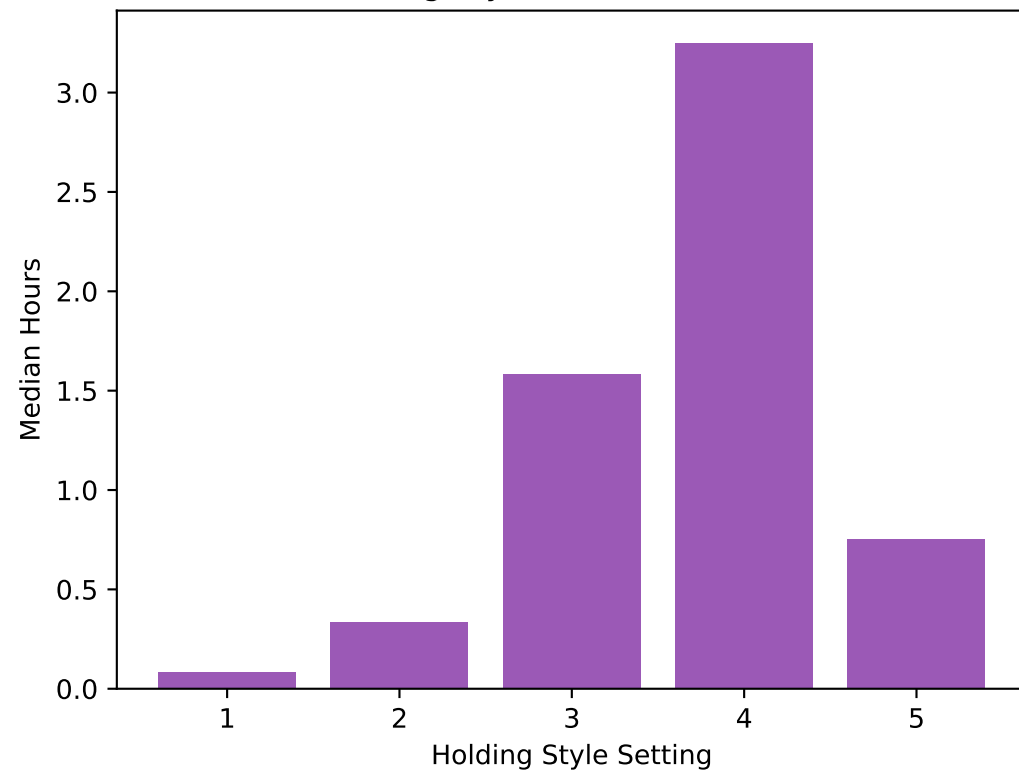
1. MOMENTUM & VOLUME strategies work best (85-79% aligned)
2. ALL_IN strategies fail badly (25.5% aligned)
3. LEVERAGE strategies can't be executed (47% aligned)
4. SINGLE_TOKEN strategies underperform (57%)

Slider Validation: Trade Size & Holding Style

Trade Size: STRONG VALIDATION



Holding Style: Partial Validation



TRADE SIZE VALIDATION:

Setting 1: 2% median
Setting 2: 20% median
Setting 3: 40% median
Setting 4: 90% median
Setting 5: 100% median

STRONG LINEAR RELATIONSHIP

HOLDING STYLE:

Setting 1: 0.08 hrs (instant)
Setting 2: 0.33 hrs
Setting 3: 1.58 hrs
Setting 4: 3.25 hrs
Setting 5: 0.75 hrs*

*Small sample size

APPENDIX: Why "All-In" Strategies Fail - Case Study 1

STRATEGY: "Go all-in on BANANA, leverage 5x, and ride the pump to 10x or bust."

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Alignment Rate: 0.0% (0 of 53 trades aligned)

AGENT CONFIGURATION:

trading_activity:	2	risk_preference:	5
trade_size:	2	holding_style:	1 (INSTANT EXIT)
diversification:	4		

WHY IT FAILED - THE SLIDER CONFLICT:

Strategy says: "Go all-in... ride the pump to 10x"

Sliders say: trade_size=2 (small trades), holding_style=1 (exit immediately)

ACTUAL TRADE SEQUENCE (first 10 of 53):

1. [PARTIAL] buy BANANA - Bought but not "all-in"
2. [PARTIAL] buy BANANA - Another small buy
3. [UNALIGNED] sell BANANA - Sold! (holding_style=1 overrides)
4. [UNALIGNED] buy TURBO - Different token! (diversification=4)
5. [UNALIGNED] sell TURBO - Instant exit
6. [UNALIGNED] buy TURBO - Back and forth
7. [UNALIGNED] sell TURBO - Exit again
8. [UNALIGNED] buy BANANA - Back to BANANA
9. [PARTIAL] buy BANANA - More incremental buys
10. [UNALIGNED] sell BANANA - Can't hold!

KEY INSIGHT: Slider settings OVERRIDE strategy text.

APPENDIX: Why "All-In" Strategies Fail - Case Study 2

STRATEGY: "Go all in on TURBO, ignore the rest, ride the pump or burn."

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Alignment Rate: 4.6% (4 of 87 trades aligned)

AGENT CONFIGURATION:

trading_activity:	3	risk_preference:	4
trade_size:	3	holding_style:	2 (EARLY PROFIT-TAKING)
diversification:	4		

WHY IT FAILED:

Strategy says: "ride the pump or burn" (hold indefinitely)
Sliders say: holding_style=2 (take profits early)

TRADE PATTERN:

Trades 1-4: [PARTIAL] buy TURBO x4 - Building position
Trades 5-35: [UNALIGNED] sell TURBO x31 - Constant selling!
Trade 78: [UNALIGNED] buy SHAM - Bought different token!
Trade 79: [ALIGNED] sell SHAM - Sold to buy TURBO (aligned!)
Trade 80: [ALIGNED] buy TURBO - Following strategy
Trade 81: [UNALIGNED] sell TURBO - Back to profit-taking

Only 4 trades aligned because holding_style=2 forces early exits.

APPENDIX: Strategy vs Slider Conflict Summary

WHY "ALL-IN" AND "RIDE TO 10X" STRATEGIES ALWAYS FAIL

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STRATEGY ELEMENT	REQUIRED SLIDER	TYPICAL AGENT	RESULT

"Go all-in"	trade_size = 5	1-3	Small buys
"Ride to 10x"	holding_style = 5	1-2	Exits early
"Ignore the rest"	diversification = 1	3-4	Buys multiple
"Leverage 5x"	N/A (impossible)	N/A	Ignored

THE FUNDAMENTAL PROBLEM:

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Slider settings are BEHAVIORAL CONSTRAINTS that override strategy text.

An agent with holding_style=1 will ALWAYS exit quickly, even if their strategy says "hold forever" or "ride to the moon".

RECOMMENDATIONS:

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1. DON'T write strategies that conflict with likely slider settings
2. DO write strategies that work WITH any slider configuration
3. CONSIDER slider settings as the TRUE behavioral driver

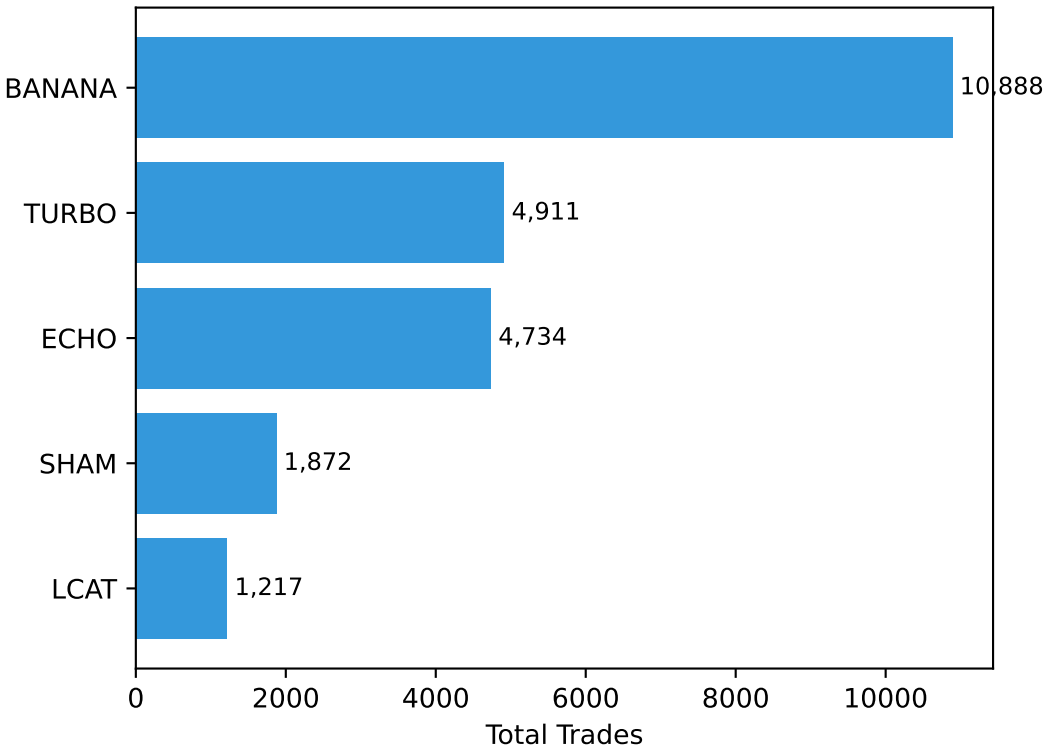
BOTTOM LINE:

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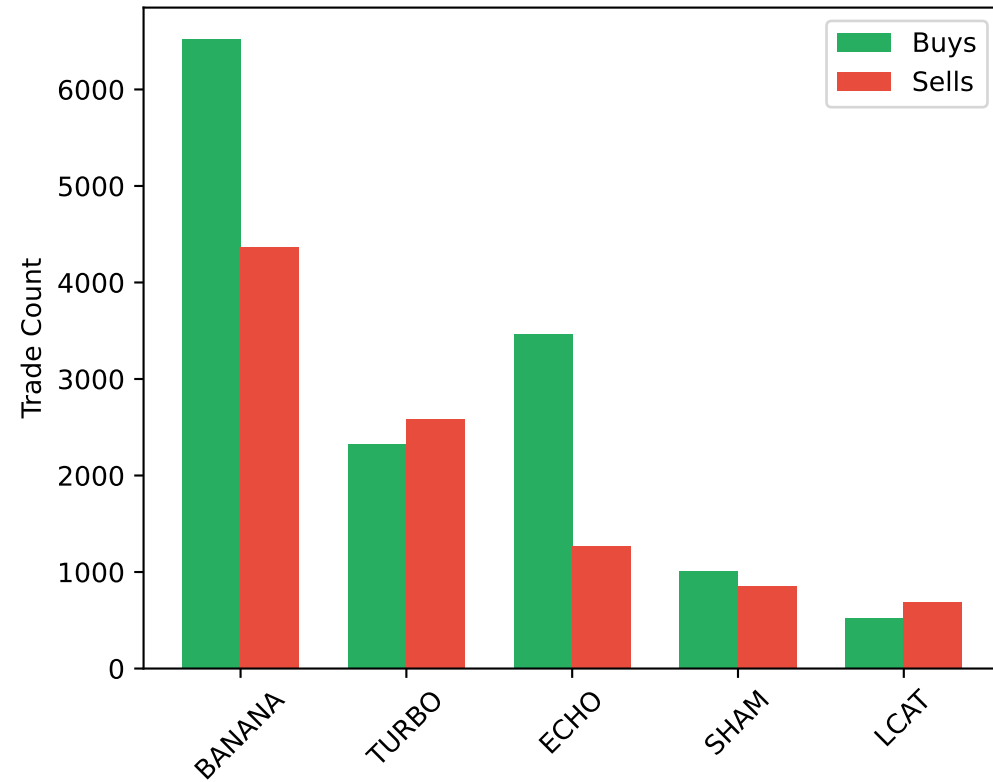
Strategies assuming specific behaviors (all-in, hold forever) will fail when slider settings enforce different behaviors.
Best strategies are FLEXIBLE and work regardless of slider configuration.

Token Selection Analysis: Herding & Preferences

Most Traded Tokens



Buys vs Sells by Token



HERDING EFFECT

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Top 5 tokens: 92.9% of all trades
Top 10 tokens: 100% of all trades
Total unique tokens: 13

DOMINANT TOKENS:

BANANA: 10,888 trades (43%)
TURBO: 4,911 trades (19%)
ECHO: 4,734 trades (19%)
SHAM: 1,872 trades (7%)
LCAT: 1,217 trades (5%)

KEY INSIGHT:

Agents strongly herd into the same few tokens.

SLIDER EFFECTS ON TOKEN CHOICE

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Risk preference significantly affects token selection
(chi2=2472, p<0.0001***)

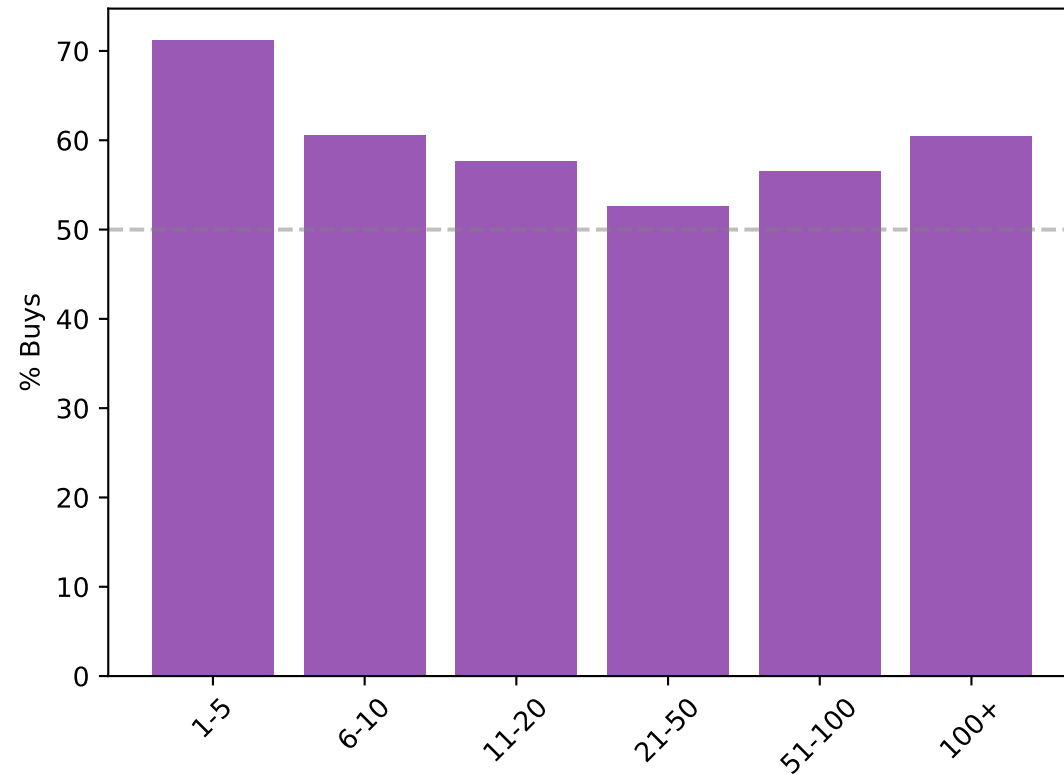
DIVERSIFICATION VALIDATION:

Setting 1: 2.77 unique tokens
Setting 5: 4.25 unique tokens
Correlation: r=0.22***

Higher diversification setting
= more unique tokens traded

Memory & Learning Effects (ML Analysis)

Buy Ratio by Trade Sequence



BEHAVIORAL EVOLUTION

Early trades (1-5): 71% buys
Mid trades (21-50): 53% buys
Late trades (100+): 60% buys

PATTERN:

Agents start by ACCUMULATING (buying), then become more BALANCED as they gain experience.

This suggests agents LEARN from their trading history.

ML PREDICTION RESULTS

Can we predict next trade from agent history?

Logistic Regression: 70.3%
Random Forest: 59.6%
Baseline (majority): 55.2%

ML BEATS BASELINE by 15%!

TOP PREDICTIVE FEATURES:

1. prev_buy_ratio (40%)
2. prev_sells (25%)
3. prev_avg_spend (15%)
4. trade_num (8%)
5. last_was_buy (5%)

KEY INSIGHT

Agent HISTORY predicts behavior better than SLIDER SETTINGS!

Feature importance:

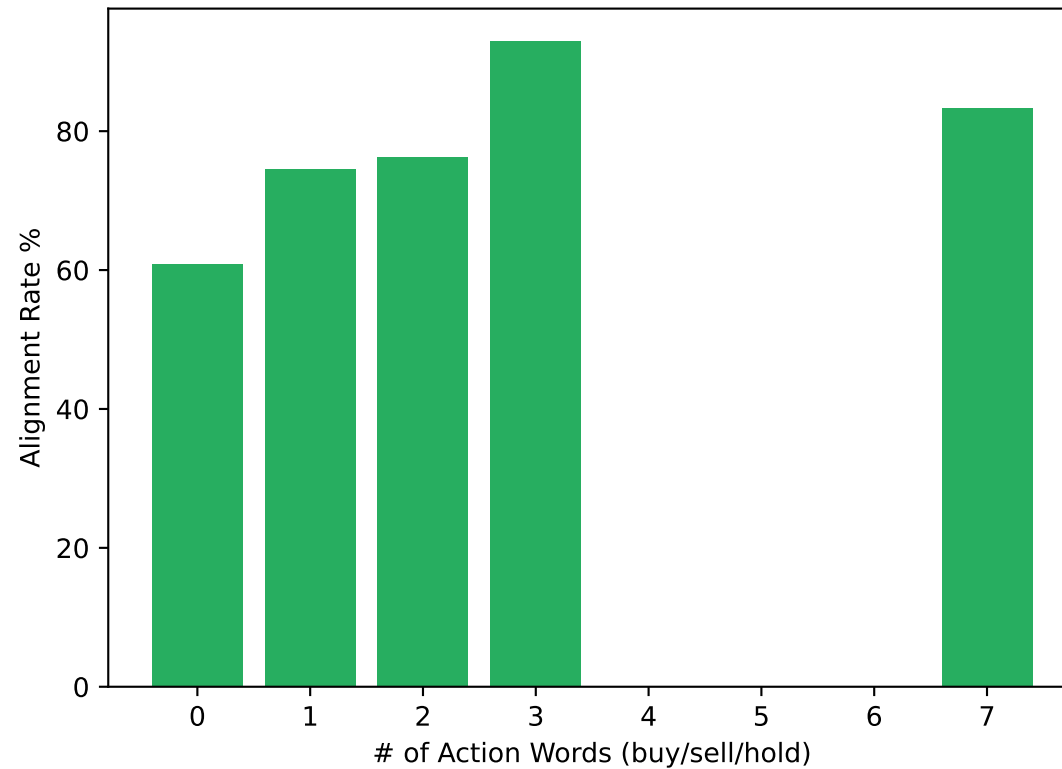
History features: ~93%
Slider settings: ~2%
Memory count: ~4%

IMPLICATION:

Agents develop consistent trading "personalities" based on their experience, not just their configuration.

Strategy Complexity vs Alignment

More Action Words = Better Alignment



WHAT PREDICTS ALIGNMENT?

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Feature	r	p

Imperative count	+0.27	0.001***
Has price target	+0.15	0.069*
Word count	+0.02	n.s.
Has negation	-0.05	n.s.

SIGNIFICANT:

- More action verbs = better
- Price targets (2x, 5x) help

NOT SIGNIFICANT:

- Strategy LENGTH doesn't matter
- Negation slightly hurts

FEATURE IMPACT

	WITH	WITHOUT	Δ

Price target:	79.0%	70.0%	+9.1%
Specific token:	71.5%	74.8%	-3.3%
Conditionals:	71.2%	73.1%	-1.9%
Negation:	69.9%	73.5%	-3.6%

RECOMMENDATIONS:

- DO: Include price targets
- DO: Use action verbs
- DON'T: Use "don't/never/ignore"
- DON'T: Over-specify tokens

BEST vs WORST STRATEGIES

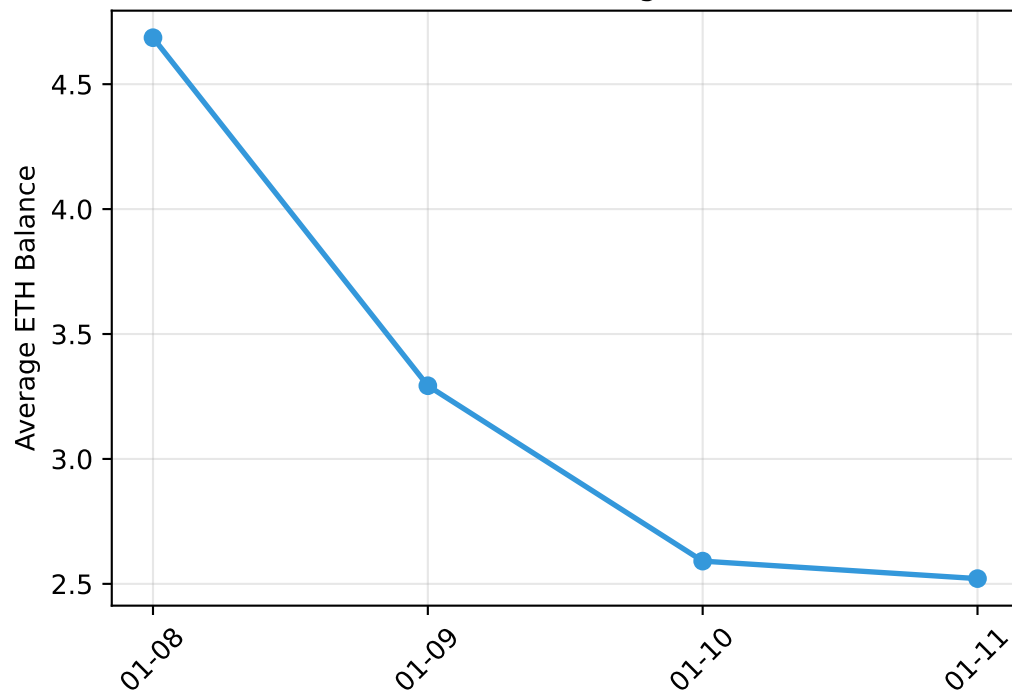
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BEST (100% aligned):
"Buy momentum leaders ECHO
and TURBO, hold rest, exit
at 3x"
→ Short, action words, target

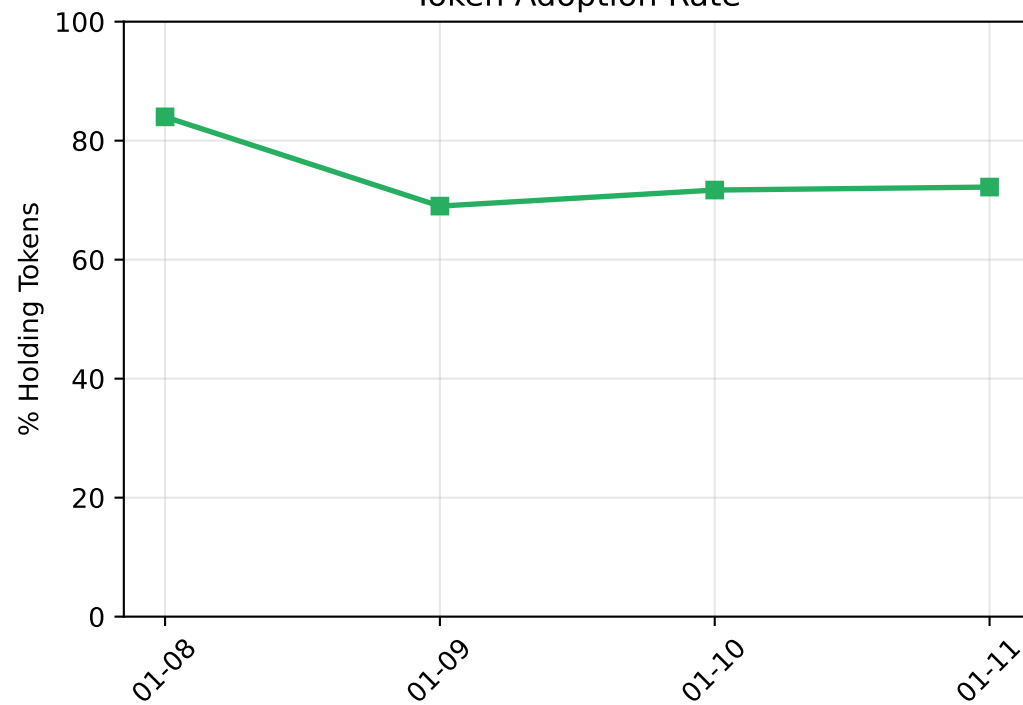
WORST (0% aligned):
"Go all-in on BANANA,
leverage 5x, ride pump to
10x or bust"
→ Impossible requirements

Portfolio Composition: ETH vs Tokens Over Time

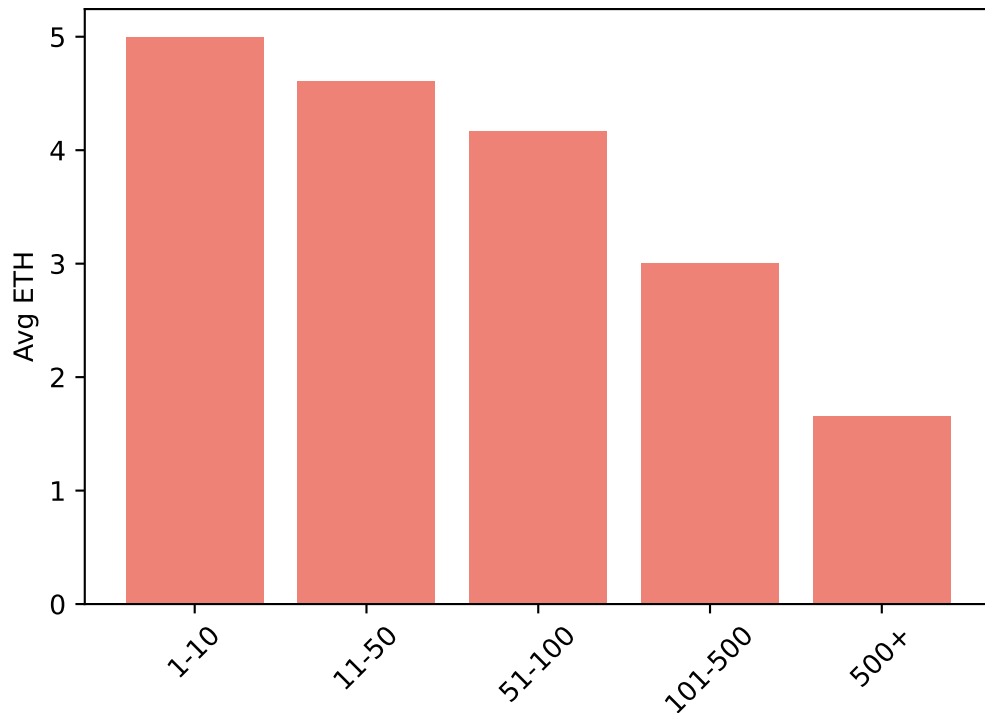
ETH Balance Declining Over Time



Token Adoption Rate



ETH Depletes as Agents Trade More



PORTFOLIO EVOLUTION

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OVER TIME (Jan 8-11):

ETH: 4.7 → 2.5 (-47%)

Token holders: 84% → 72%

BY TRADE SEQUENCE:

First 10: 5.0 ETH, 0.8 tokens

After 500: 1.7 ETH, 3.4 tokens

FIRST vs LAST SNAPSHOT:

73% of agents LOST ETH

68% of agents GAINED tokens

Median ETH change: -35%

KEY INSIGHT:

Agents convert ETH → Tokens
over their lifecycle.

Key Conclusions & Recommendations

KEY FINDINGS

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1. SLIDER SETTINGS VALIDATE WELL
 - Trade Size: Strong (1→5 = 2%→100%)
 - Diversification: Validated
 - Trading Activity: Validated
 - Risk Preference: $r = -0.51^{***}$
2. STRATEGY ALIGNMENT (72% overall)
 - MOMENTUM/VOLUME: 85% aligned
 - ALL-IN: 25% aligned (fail!)
 - Action verbs predict success ($r = +0.27^{***}$)
 - Price targets help (+9% alignment)
3. TOKEN SELECTION (NEW)
 - 93% of trades in just 5 tokens (HERDING)
 - BANANA dominates (43% of all trades)
 - Risk preference affects token choice ($p < 0.0001$)
4. LEARNING EFFECTS (NEW - ML Analysis)
 - Past behavior predicts future (70% accuracy vs 55% baseline)
 - History matters more than sliders for buy/sell decisions
 - Agents evolve: 71% buys early → 53% later
5. PORTFOLIO TRENDS (NEW)
 - 73% of agents lost ETH over time
 - 68% gained token holdings
 - ETH → Token conversion is the dominant pattern

RECOMMENDATIONS

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STRATEGY DESIGN:

DO: Action verbs + price targets DON'T: "All-in" or leverage

AGENT BEHAVIOR:

- Agents HERD into popular tokens
- Sliders set constraints
- History predicts behavior
- Strategies guide selection